Name ____ Date ___________ Period _____

Parent Functions I: √x and |x|

Algebra II Honors | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph and analyze the square root and absolute value parent functions, write absolute value as a piecewise rule, and explain why the two behave differently for negative inputs.

Vocabulary

Reflectional symmetry — the graph can be _____________ across a line and land exactly on itself.

x-intercept(s) — where the graph crosses the x-axis, where y = _____. **y-intercept(s)** — where x = _____.

Zeros — the x-value(s) that make a function equal _____ — the same locations as the x-intercepts.

Parent function — the simplest member of a family, with nothing added or multiplied.

Piecewise rule — one function defined by ______________________ on different parts of the domain.

Principal square root — √a means the _______________ root only, which is why √x is a function.

Part 1 — Build Both Tables

Fill both. Note where outputs are undefined and where they repeat.

x	-4	-1	0	1	4	9
f(x) = √x						

x	-4	-1	0	1	4	9
g(x) = |x|						

One table has undefined entries, the other has repeats. Explain what each tells you:

____

____

√x has a restricted _____________; |x| is not _______________, which will matter on Day 12.

Part 2 — Absolute Value as a Piecewise Rule

Absolute value is distance from zero, so it never returns a negative. Written as two cases:

|x| = _____________ when x ≥ 0, and |x| = _____________ when x < 0.

The second case looks like it produces a negative. Explain why it does not:

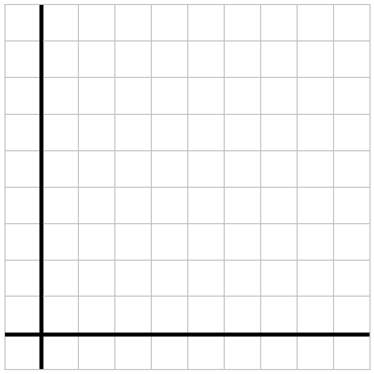
____

If x is already negative, ______________________. For example -(-5) = 5.

Part 3 — Graph Them Both

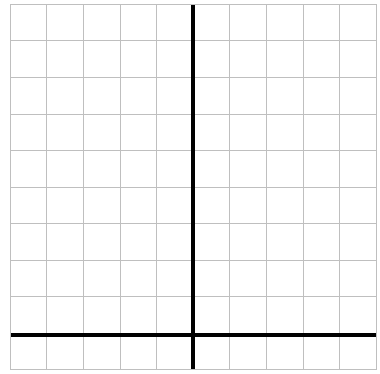
Each square is 1 unit. Plot from your tables, then connect smoothly.

A. $f(x) = \sqrt{x}$ x from 0 to 9



Domain _____________ Range _____________

B. $g(x) = |x|$ x from -5 to 5



Domain _____________ Range _____________

Part 4 — Solution Counts

Solve each. The number of answers is the point, not just the answers.

1. $|x| = 6$ → x = _______________________ how many? _________
2. $\sqrt{x} = 6$ → x = __________________ how many? _________
3. $|x| = 0$ → x = _______________________ how many? _________
4. $|x| = -3$ → _______________________

Item 4 has no solution. What does that say about the range of $|x|$? __________________________________

Connect item 1 to the graph: why do two x-values give the same output?

____

The V is symmetric about the _____________, so a horizontal line meets it twice.

Part 5 — Going Further

1. Give the domain of each, in interval notation.

$\sqrt{x - 7}$ __________________ $\sqrt{3x + 6}$ __________________ $|x - 7|$ __________________

2. Why does the absolute value one have no restriction while the radicals do?

____

3. On the interval [1, 9], give the minimum and maximum of $\sqrt{x}$. min ___________ max ___________

4. Is $\sqrt{x^2}$ the same function as $|x|$? Test x = -5 and x = 5, then decide.

____

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Yes — ____

Exit Ticket

Both parent functions have minimum 0 and pass through the origin.

Name one attribute that separates them: ___________________________ and one they share besides those two: _________________________

Write $|x|$ as a piecewise rule from memory: ______________________________________